E-Commerce Performance Analytics

SQL Server | Python | Power BI | Generative AI

End-to-End Sales, Customer Funnel & Marketing Performance Report

Dataset Period	Oct 2019
Records Analyzed	160,000 clickstream events
Prepared By	Swati
Tools Used	SQL Server, Python (Pandas/Matplotlib), Power BI, Gemini AI

1. Project Overview

This project delivers an end-to-end analytics solution for an e-commerce business, covering the full pipeline from raw clickstream data to executive-ready insights. The workflow spans SQL Server for data storage and KPI extraction, Python for data cleaning and exploratory analysis, Power BI for interactive dashboarding, and a generative AI layer for automated executive summarization.

The underlying dataset captures 160,000 customer interaction events (Views, Cart additions, and Purchases) recorded across October 2019, alongside a supporting promotions table linking discounted products to specific dates. Together these describe the full customer journey — from product discovery through to checkout — and the effect of promotional activity on that journey.

1.1 Objectives

- Consolidate and clean raw transactional data using SQL Server.
- Quantify sales performance across brand, category, geography, and time.
- Analyze the customer purchase funnel (View → Cart → Purchase) to identify drop-off points.
- Evaluate the effectiveness of promotional discounting on revenue.
- Visualize findings in an interactive Power BI dashboard for business stakeholders.
- Automate executive summary generation using a generative AI model fed directly from SQL-derived KPIs.

1.2 Tools & Tech Stack

Layer	Tool	Purpose
Data Storage & Querying	Microsoft SQL Server (T-SQL)	Data staging, cleaning views, KPI aggregation views
Data Processing	Python (Pandas, NumPy)	Data cleaning, merging, feature engineering
Exploratory Analysis	Python (Matplotlib)	Funnel, revenue, and pricing visualizations
Business Intelligence	Microsoft Power BI	Interactive dashboard for stakeholders
AI Summarization	Google Gemini API	Automated narrative executive summary from KPIs

2. Data Overview & Methodology

Two source tables were loaded into SQL Server: **Sales_Data_Ecommerce** (160,000 rows of clickstream events with user, product, pricing, geography, and timestamp attributes) and **Promotion** (78 records mapping discounted products to specific dates). Data quality checks in SQL and Python confirmed no missing values and duplicate records were removed prior to analysis. The two tables were joined on product ID and event date to flag which purchase events occurred under an active promotion.

Table	Rows	Key Fields
Sales_Data_Ecommerce	160,000	user_id, event_date, event_type, product_id, category, brand, price, State, Chann
Promotion	78 (after cleaning)	Promotion_Id, Date, Discount, ProductId

The product catalog spans **10,124 unique products** across **12 categories** and **50 brands**, sold through two channels (Browser and App) into all 50 US states.

3. Executive KPI Summary

Total revenue reached **$5,650,977.17** across **14,647 orders**, an average order value of **$385.81**. The platform recorded **144,335** unique customers, of whom only **13,914 (9.6%)** completed a purchase — the single largest opportunity area identified in this analysis.

$5.65M TOTAL REVENUE	**14,647** TOTAL ORDERS	**$385.81** AVG ORDER VALUE
11.63% CONVERSION RATE	**75.39%** CART→PURCHASE	**13,914** PURCHASING CUST.

Metric Group	KPI	Value
Financials	Total Revenue	**$5,650,977.17**
	Total Orders	**14,647**
	Average Order Value	**$385.81**
	Highest Discount Offered	**35%**
Customers & Catalog	Total Customers	**144,335**
	Purchasing Customers	**13,914 (9.64%)**
	Total Products / Categories / Brands	**10,124 / 12 / 50**
Funnel	Total Views / Carts / Purchases	**125,925 / 19,428 / 14,647**
	Cart Rate (Cart / View)	**15.43%**
	Cart-to-Purchase Rate	**75.39%**
	Overall Conversion Rate	**11.63%**
Key Drivers	Top Brand	**Apple**
	Top Category	**Electronics**
	Top State	**Mississippi (MS)**
	Best Sales Day / Hour	**Sunday / 10:00 AM**
	Best Channel	**Browser**

4. Sales Performance

Revenue is heavily concentrated: **Apple** alone generated over $3.0M — more than double the next-largest brand, Samsung. At the category level, **Electronics** contributed roughly 88% of total revenue, dwarfing Appliances and Computers, the next-largest categories.

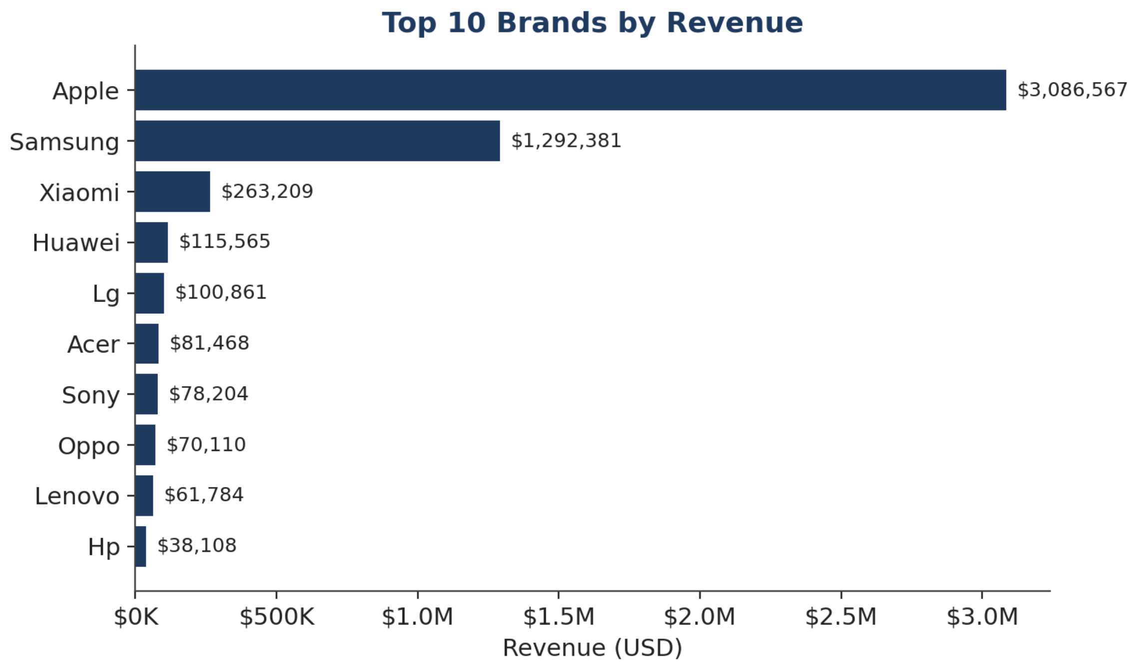

Figure 1: Top 10 brands by revenue — Apple leads by a wide margin.

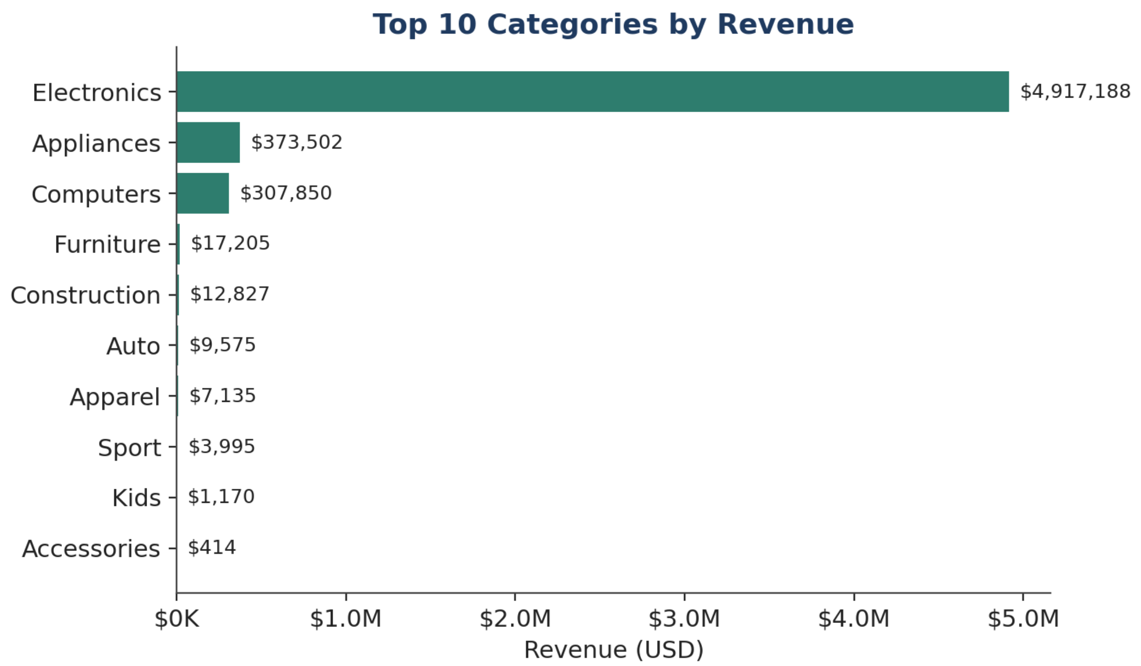

Figure 2: Top 10 categories by revenue — Electronics dominates the mix.

Geographically, revenue is well-balanced across the top 10 states (ranging $119K–$134K each), with Mississippi narrowly leading. Time-based analysis shows a clear morning peak: revenue is highest between 8–11 AM, and Sunday is the strongest day of the week.

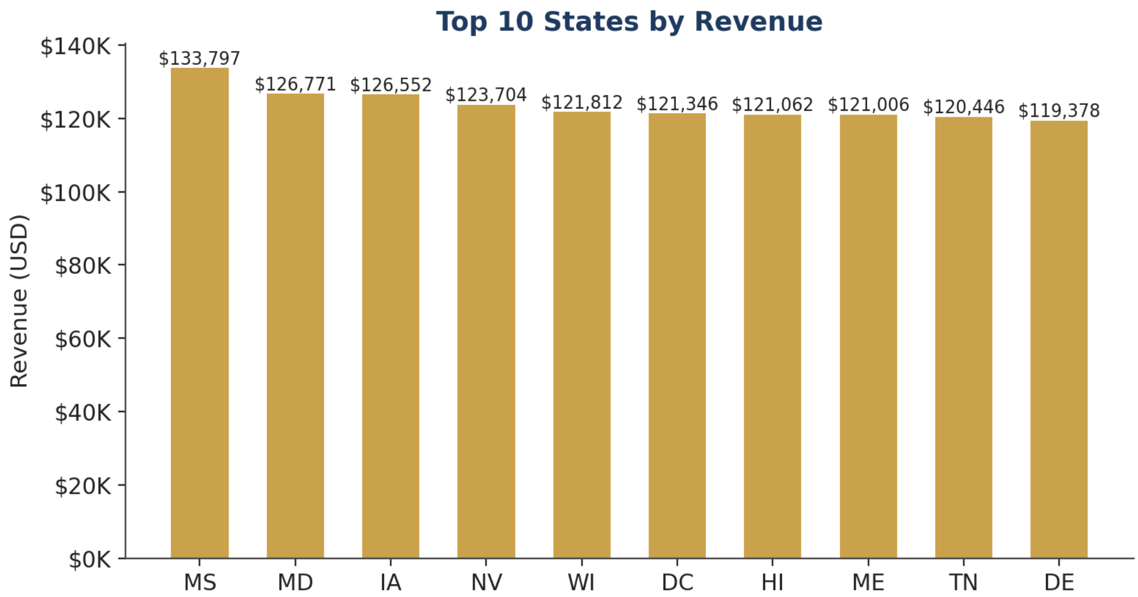

Figure 3: Top 10 states by revenue — a relatively even geographic spread.

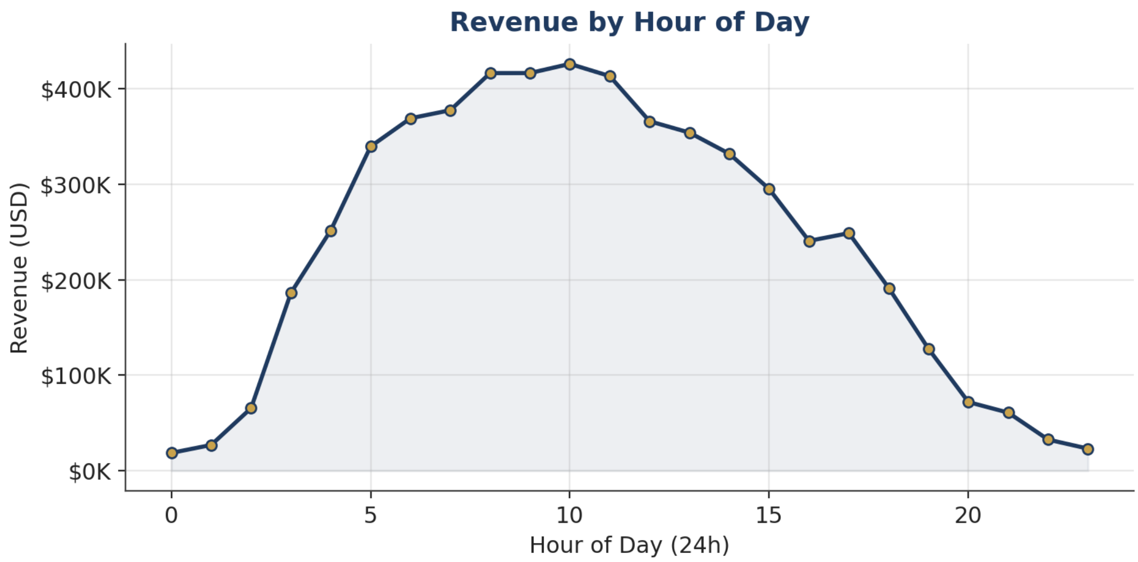

Figure 4: Revenue by hour of day — morning hours (8–11 AM) are peak trading hours.

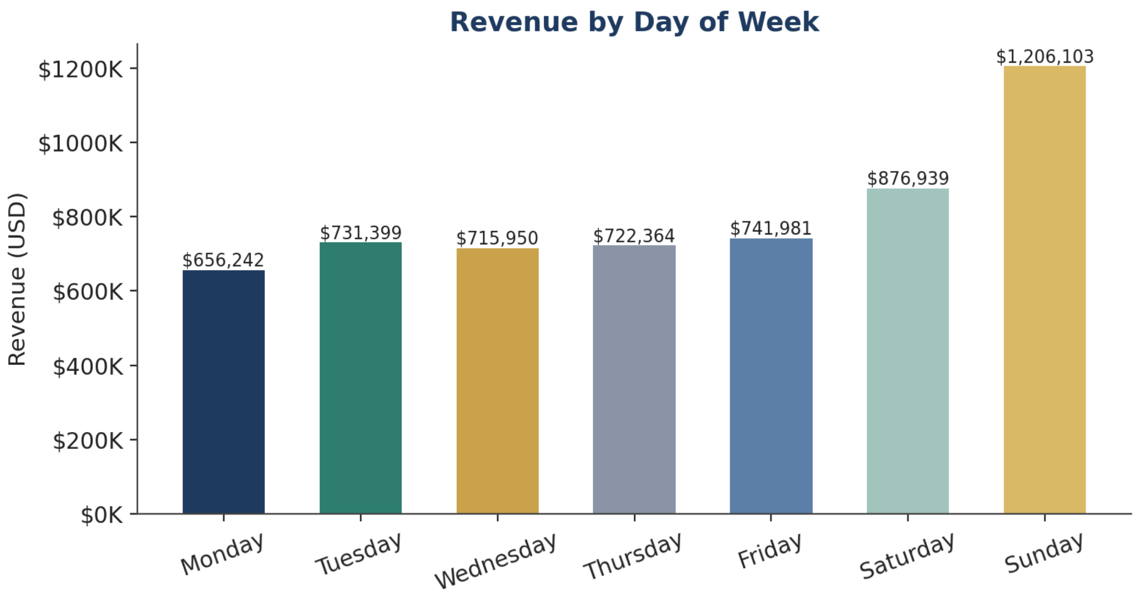

Figure 5: Revenue by day of week — Sunday is the strongest trading day.

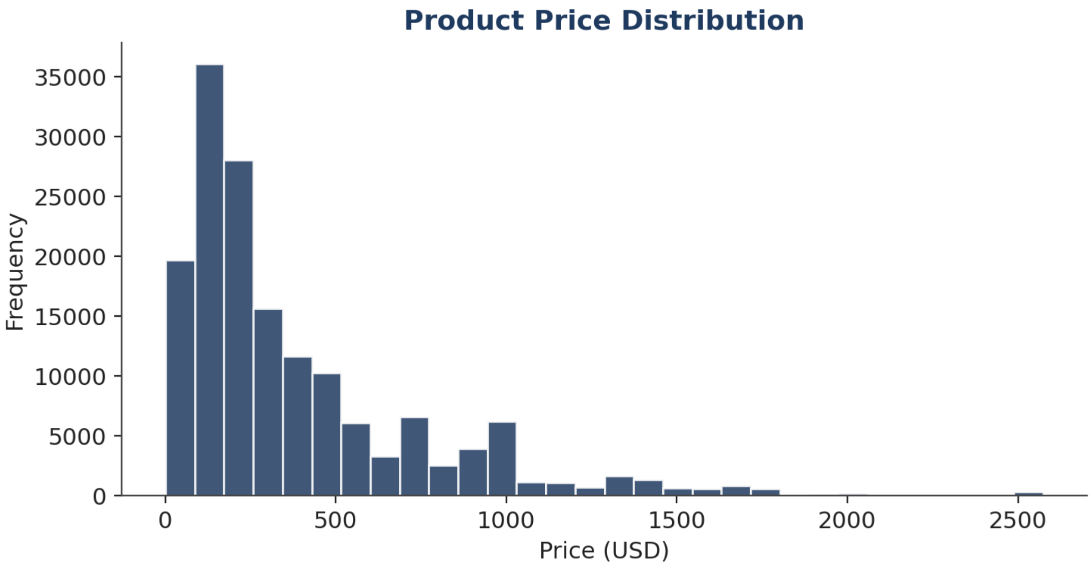

Figure 6: Product price distribution — most transactions are under $300; premium-priced items ($1,500+) are rare.

5. Customer Purchase Funnel

Of 125,925 recorded product views, 19,428 (15.43%) progressed to a cart addition, and 14,647 of those carts (75.39%) converted to a completed purchase — giving an overall View-to-Purchase conversion rate of 11.63%. The largest drop-off in the funnel occurs at the top: **View** → **Cart**, not at checkout. Once a product is in the cart, the business converts it into a sale at a strong rate.

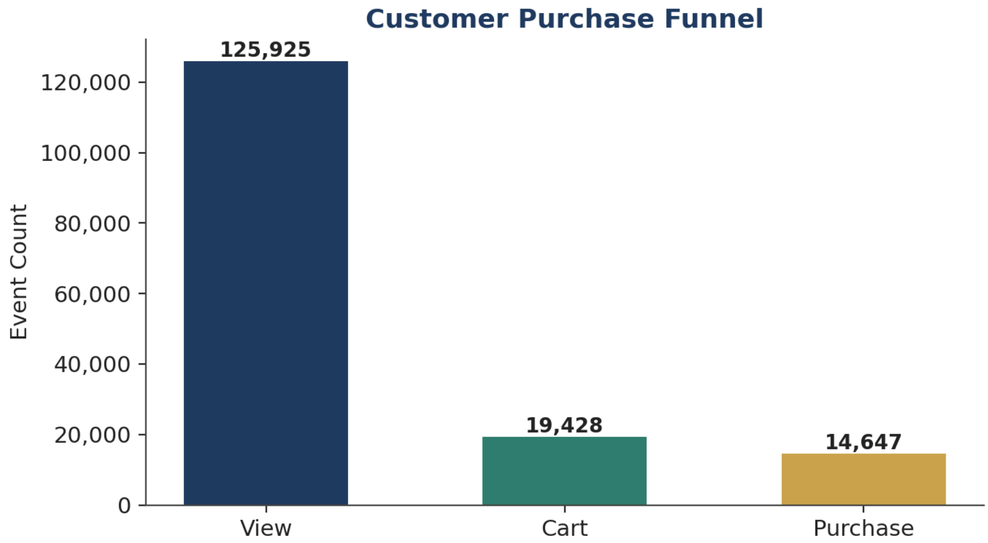

Figure 7: Customer purchase funnel from View through to Purchase.

5.1 Customer Activation Gap

Out of 144,335 total customers tracked in the system, only 13,914 (9.64%) have completed a purchase. This means over 90% of the customer base is registered or has interacted with the platform but has never converted — representing the single largest growth lever identified in this analysis.

6. Marketing & Promotion Analysis

Promotional activity in the dataset is minimal in scope: only 78 product-date combinations carried an active discount (up to a maximum of 35%). Of total revenue, just **$23,103** (about 0.4%) was generated under an active promotion, with the remaining **$5.63M** (99.6%) coming from full-price sales. This indicates the business is not currently dependent on discounting to drive volume — but it also means promotions, as currently structured, are too limited in reach to meaningfully move overall revenue.

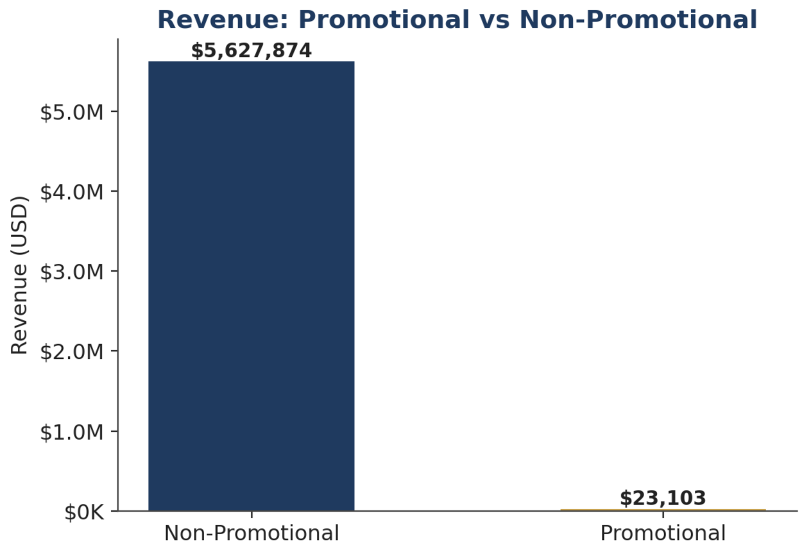

Figure 8: Revenue generated under active promotions vs. full-price sales.

The **Browser** channel outperforms the App channel as the primary source of purchase revenue, suggesting checkout and browsing experience investment should be prioritized on desktop/web first.

7. Business Risks

Customer database under-activation. Over 90% of the 144,335 tracked customers have never purchased, pointing to high acquisition cost relative to realized conversion, or friction early in the funnel.

Brand & category concentration. Apple and Electronics together account for the large majority of revenue. Any supply disruption, pricing change, or shift in consumer tech spending would disproportionately affect the business.

Channel concentration. The Browser channel drives the majority of purchase revenue; changes to browser-based tracking or a shift in customer device preference could disrupt sales if the App experience is not competitive.

Underutilized promotions. Promotions currently touch a very small share of the catalog and contribute under 0.5% of revenue, meaning discounting is not yet being used as a lever to activate dormant customers or clear inventory.

8. Recommendations

1. Activate the silent customer base. Launch targeted email/SMS remarketing to the ~130,000 non-purchasing registered customers, using a welcome incentive (within the proven ≤35% discount ceiling) to drive first-time conversion.

2. Focus growth effort on top-of-funnel. Since Cart-to-Purchase conversion is already strong (75.4%), prioritize investment in driving more View-to-Cart actions — e.g. product page optimization, recommendation widgets, and urgency messaging — rather than further checkout optimization.

3. Time campaigns to peak windows. Align push notifications, newsletters, and promotional launches with the Sunday, 8–11 AM peak-trading window to maximize reach and conversion.

4. Diversify beyond Apple & Electronics. Use high-traffic Apple/Electronics product pages to cross-sell the other 11 categories and 49 brands via "frequently bought together" placements, reducing concentration risk and lifting average order value.

5. Expand promotional reach deliberately. Test broader (but still capped) discounting on a larger slice of the catalog to measure incremental impact on both new-customer activation and existing full-price cannibalization.

9. Deliverables

- SQL Server database with cleaned tables and reusable analytical views (Executive KPIs, Revenue by Brand/Category/State, Customer Funnel, Promotion Performance).
- Python EDA notebook covering data cleaning, feature engineering, and exploratory visualizations.
- Interactive Power BI dashboard for stakeholder self-service exploration.
- AI-assisted executive summary generation pipeline, sourcing KPIs directly from the SQL Server view to minimize manual reporting effort.

10. Conclusion

This project demonstrates a complete, reproducible analytics workflow — from raw SQL Server data to a polished, decision-ready report — covering sales performance, funnel efficiency, and promotional impact. The business is fundamentally healthy, with a highly efficient conversion funnel once a customer reaches the cart stage. The clearest paths to growth are activating the large base of non-purchasing customers and reducing reliance on a single brand and category for the majority of revenue.